

MKM-EF-001 — λ Sensitivity Analysis (one-off)

Component: Event Frequency Model — MKM-EF-001 (governance record v1.1.0) **Purpose:** A dedicated study of the model's sensitivity to the Poisson arrival rate λ — its single core uncertainty — propagating a plausible λ band both deterministically and probabilistically through to the priced PRS spread. **Catchment / book:** halong · 200 properties (188 with flood exposure) · seed $\lambda = 4.5$ events/yr. **Date:** 2026-07-28 · **Reproduce:** models.frequency.rate_sensitivity, lambda_price_distribution over the halong propertyhc conditionals.

1. Why λ is the analysis that matters

A PRS severe-trigger spread is $1 - \exp(-\lambda \cdot p)$, so λ is a **linear multiplier** on the annual hazard. The prior sensitivity report established that the spread tracks λ almost one-for-one and that the distributional form and Monte-Carlo sampling are second-order. What that report also established is the open item: **λ 's level is unvalidated on a synthetic catchment** — the gauge record is generated from an assumed frequency and the per-gauge extraction recovers its own target, so both validation routes are circular (plan §5).

λ therefore carries genuine uncertainty, and because it is a linear multiplier that uncertainty flows straight into every quoted spread. This study quantifies exactly how much, on the real book, over a **$\pm 50\%$ plausible band on λ** (2.25–6.75 /yr).

2. Deterministic transmission — the $\pm 50\%$ fan on the real book

Applying each λ shock to every property's fitted per-event conditional and re-pricing $1 - \exp(-\lambda \cdot p)$, then averaging across the 200-property book (base average **198.3 bps**, matching the port):

λ shock	λ (/yr)	book avg spread	change
×50%	2.25	99.9 bp	−49.6%
×80%	3.60	159.1 bp	−19.7%
×100%	4.50	198.3 bp	—
×120%	5.40	237.1 bp	+19.6%
×150%	6.75	295.0 bp	+48.8%

The response is **near-proportional**: a given percentage error in λ moves the book's average spread by essentially the same percentage. The closed-form reason is that $\lambda \cdot p \ll 1$ at realistic conditionals, so $1 - \exp(-\lambda p) \approx \lambda p$ and the derivative $\partial \text{spread} / \partial \lambda = p \cdot \exp(-\lambda p)$ is almost constant across the band.

3. Probabilistic propagation — λ 's uncertainty becomes price uncertainty

Treating λ as uncertain with the $\pm 50\%$ band as the 5th–95th percentiles of its prior (P5 = 2.25, P50 = 4.5, P95 = 6.75) and propagating through the monotonic map $\text{spread}(\lambda)$ — exactly, since a percentile of λ maps to the same percentile of the spread:

Book average spread

	P5	P50	P95
λ (/yr)	2.25	4.50	6.75
book avg spread	99.9 bp	198.3 bp	295.0 bp

	P5	P50	P95
vs median	–50%	—	+49%

Induced 90% credible interval on the book’s average spread: [–50%, +49%]. The **passthrough** — the ratio of the spread’s relative span to λ ’s — is **0.984** at book level (0.991 for a representative median-conditional property).

The interpretation is direct and important: **a $\pm 50\%$ uncertainty on λ is a $\pm 50\%$ uncertainty on the entire book’s PRS valuation.** The price carries the same relative model-uncertainty band as the rate, because in this regime it *is* the rate. There is no diversification of this uncertainty across the book — λ is a catchment-wide multiplier, so it shifts every asset together.

4. Where the one-for-one relation weakens

Passthrough is below 1 only through the convexity of $1 - \exp(-\lambda p)$: as $\lambda \cdot p$ grows the curve bends and the upside is compressed relative to the downside. On this book the effect is small (passthrough 0.98) because conditionals are low (median $p \approx 0.004$, so $\lambda p \approx 0.02$). It would matter only for a gauge/asset with a much higher conditional — a near-certain-flood location — where the spread saturates towards 1 and further λ has diminishing effect. No such asset is material in the halong book. So across the realistic range the one-for-one approximation is safe to reason with.

5. Conclusion and implication for validation

Finding	Value
Book base average spread	198.3 bp ($\lambda = 4.5$)
Book avg spread under $\pm 50\% \lambda$	99.9 – 295.0 bp
Induced 90% CI on the book	[–50%, +49%]
λ -to-price passthrough	0.98 ($\approx$ one-for-one)
Diversifiable across the book?	No — λ is a catchment-wide multiplier

The model’s exposure to the Poisson rate is now fully characterised: **the PRS book inherits λ ’s uncertainty essentially one-for-one and undiversified.** A $\pm 50\%$ band on λ — entirely plausible for an unvalidated rate — is a $\pm 50\%$ band on the book’s PRS valuation.

This makes the **validation of λ ’s level the single highest-value control on the model**, worth more than any refinement of the conditional, the distributional family, or the sampling. Validation requires a threshold anchored *outside* the rate — bankfull discharge or a published flood-frequency curve — and therefore **real gauge data**; it cannot be done on the synthetic catchment (plan §5). Until then, every PRS quote should be understood to carry a model-uncertainty band from λ alone of roughly the same width as the assumed uncertainty on λ .

References

- Pricing identity & annualisation: `models.frequency.annualise`; plan §4.
- λ sensitivity engine: `models.frequency.sensitivity(rate_sensitivity, lambda_price_distribution)`.
- λ circularity / why the level is unvalidated: plan §5.
- Prior overview: `ef001_sensitivity_report.md`. Governance: MKM-EF-001 v1.1.0.